

XOI 5TH Training

Editorial Find the Base

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Problem Source: La Salle-Pui Ching
Programming Challenge 2019 (Problem F)

<https://lscct.com/challenge/2019.php>



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Problem Description

- Given an array of N integers $A[i]$ ($1 \leq A[i] \leq 10^5$)
 - Among base 2 to base B , figure out the base which maximizes the number of integers fulfilling the requirement.
 - Requirement: digit sum of the number (when expressed in base K) is equal to M .
-
- 1.5 seconds, 256 megabytes



Brute Force Solution?

Enumerate all bases!

Algorithm 1 Brute Force

```
1: for  $K \leftarrow 2$  to  $B$  do
2:    $countNumbers \leftarrow 0$ 
3:   for  $i \leftarrow 1$  to  $N$  do
4:      $temp \leftarrow A[i]$ 
5:      $countDigit \leftarrow 0$ 
6:     while  $tmp > 0$  do
7:        $countDigit \leftarrow temp \bmod b$ 
8:        $tmp \leftarrow tmp \text{ DIV } K$ 
9:     end while
10:    if  $countDigit = M$  then
11:       $countNumbers \leftarrow countNumbers + 1$ 
12:    end if
13:  end for
14:   $UPDATE(K, countNumbers)$ 
15: end for
```

Time complexity:
 $O(BN \log \max\{A[i]\})$



Then... Where to Optimize?

Alice: We notice that there are too many bases to calculate!

Bob: How about we observe some useful properties?

Alice: Bases... Hmmm... The formula for a decimal number is

$$\overline{a_n a_{n-1} a_{n-2} \dots a_1 a_0} = a_n \times 10^n + a_{n-1} \times 10^{n-1} + \dots + a_0.$$

Bob: Oh wait, but did you notice that the range of $A[i]$ is relatively **small**?

Alice: Indeed! This must be a way to solve the problem!



Next Stage:

Bob: What a time-consuming calculation! We need to enumerate every number, even for the large bases.

Alice: Oh, WAIT! For those cases with large bases, the transformed number wouldn't contain too many digits. Let's just consider the bases bigger than the square root of 10^5 for the moment. **(Why?)**

Bob: You are brilliant! Then, **the number after transformation will not have three or more digits!**

Both Alice and Bob smiled.



Let's Continue the Discussion!

Suppose we have $K=500$, $M=299$, and $A[i]=99999$

$i=0$ to $K-1$ $\Rightarrow 0$ to 499 $\Rightarrow 0$ to 299	$M-i$ $\Rightarrow 299-i$
a_1	a_2

For a base K ...

We can enumerate the first digit a_1
Then, we can easily obtain a_2 by digit sum

Let's calculate the complexity...

$$O(B \cdot B) = \text{?????????}$$

Wait, is it not optimized?



It's Actually Correct!

Alice: Is it an illusion? My program runs much faster than I expected.

--Alice tried submitting and got an **AC**—

Bob: Haha! You should never forget the **Harmonic Series**.

Let's take a step back from this problem and we will be back soon...



Harmonic Series in OI

- $1 + \frac{1}{2} + \frac{1}{3} + \dots \approx \log n + \gamma$
- Application: **Prime Sieve**
- Question: What is the time complexity?
- Well-known algorithms are **Sieve of Euler** and **Sieve of Eratosthenes**.



Sieve of Eratosthenes (埃拉托色尼篩)

Algorithm 2 Sieve of Eratosthenes

```
1:  $cntPrime \leftarrow 0$ 
2: for  $i \leftarrow 2$  to  $N$  do
3:   if  $isPrime[i] = \text{TRUE}$  then
4:      $cntPrime \leftarrow cntPrime + 1$ 
5:      $Prime[cntPrime] \leftarrow i$ 
6:      $j \leftarrow i \times 2$ 
7:     while  $j \leq N$  do
8:        $isPrime[j] \leftarrow \text{FALSE}$ 
9:        $j \leftarrow j + i$ 
10:    end while
11:  end if
12: end for
```

Time complexity:
 $O(N^2)$??



Let's Refine the Analysis!

- We notice that j moves along with i , so the cost associated with the loop can be different from a typical loop.

- **Technique:** For each prime, we enumerate its multiple.

- Time complexity: let N be the range that we find the primes, we obtain:

$$\sum \frac{N}{2} + \frac{N}{3} + \frac{N}{5} + \frac{N}{7} + \frac{N}{9} + \dots < \sum \frac{N}{1} + \frac{N}{2} + \frac{N}{3} + \frac{N}{4} + \frac{N}{5} + \dots = N \log N$$

- Hence, the time complexity is $O(N \log \log N)$

- $\pi(n) \sim \frac{n}{\ln n}$ (off-topic but classic)



Back to the problem

Alice: My program runs faster than I expected.

Bob: Did you forget the **Harmonic Series**?

- How could it be related to the problem?

Let's Continue the Discussion!

Suppose we have $K=500$, $M=299$, and $A[i]=99999$

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Algorithm 3 After Fixing Alice's Code

```
1: for  $K \leftarrow B_0$  to  $B$  do
2:    $countN \leftarrow 0$ 
3:   for  $firstDigit \leftarrow 0$  to  $MIN(K - 1, M)$  do
4:      $secondDigit \leftarrow M - firstDigit$ 
5:     Bob had added something here before Alice's submission...
6:
7:     if  $secondDigit > 0$  AND  $secondDigit < K$ 
8:       AND  $firstDigit \times K + secondDigit < Range + 1$ 
9:          $countNumbers \leftarrow countNumbers + c[firstDigit \times K + secondDigit]$ 
10:      end if
11:   end for
12:    $UPDATE(K, countNumbers)$ 
13: end for
```

$$O(B \times \min(K-1, B))$$



Algorithm 3 After Fixing Alice's Code

```
1: for  $K \leftarrow B_0$  to  $B$  do
2:    $countN \leftarrow 0$ 
3:   for  $firstDigit \leftarrow 0$  to  $MIN(K - 1, M)$  do
4:      $secondDigit \leftarrow M - firstDigit$ 
5:     if  $firstDigit \times K + secondDigit > maxA$  OR  $secondDigit < 0$  then
6:       BREAK
7:     if  $secondDigit > 0$  AND  $secondDigit < K$ 
8:       AND  $firstDigit \times K + secondDigit < Range + 1$ 
9:        $countNumbers \leftarrow countNumbers + c[firstDigit \times K + secondDigit]$ 
10:    end if
11:  end for
12:   $UPDATE(K, countNumbers)$ 
13: end for
```

$$\frac{A_{range}}{B_0} + \frac{A_{range}}{B_0+1} + \dots + \frac{A_{range}}{B} \\ \approx O(A_{range} \log B)$$



Back to the problem

Alice: My program runs faster than I expected.

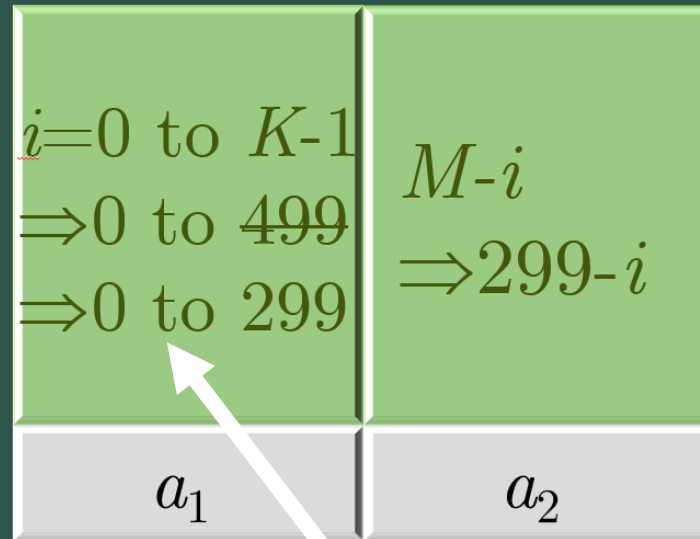
Bob: Did you forget the **Harmonic Series**?

- How could it be related to the problem?

$$\frac{A_{\text{range}}}{B_0} + \frac{A_{\text{range}}}{B_0+1} + \dots + \frac{A_{\text{range}}}{B} \\ \approx O(A_{\text{range}} \log B)$$

Let's Continue the Discussion!

Suppose we have $K=500$, $M=299$, and $A[i]=99999$



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- $A = a_1 \times K + a_2$, then what is the range of a_1 ?
- 0 to $\frac{A}{K}$!



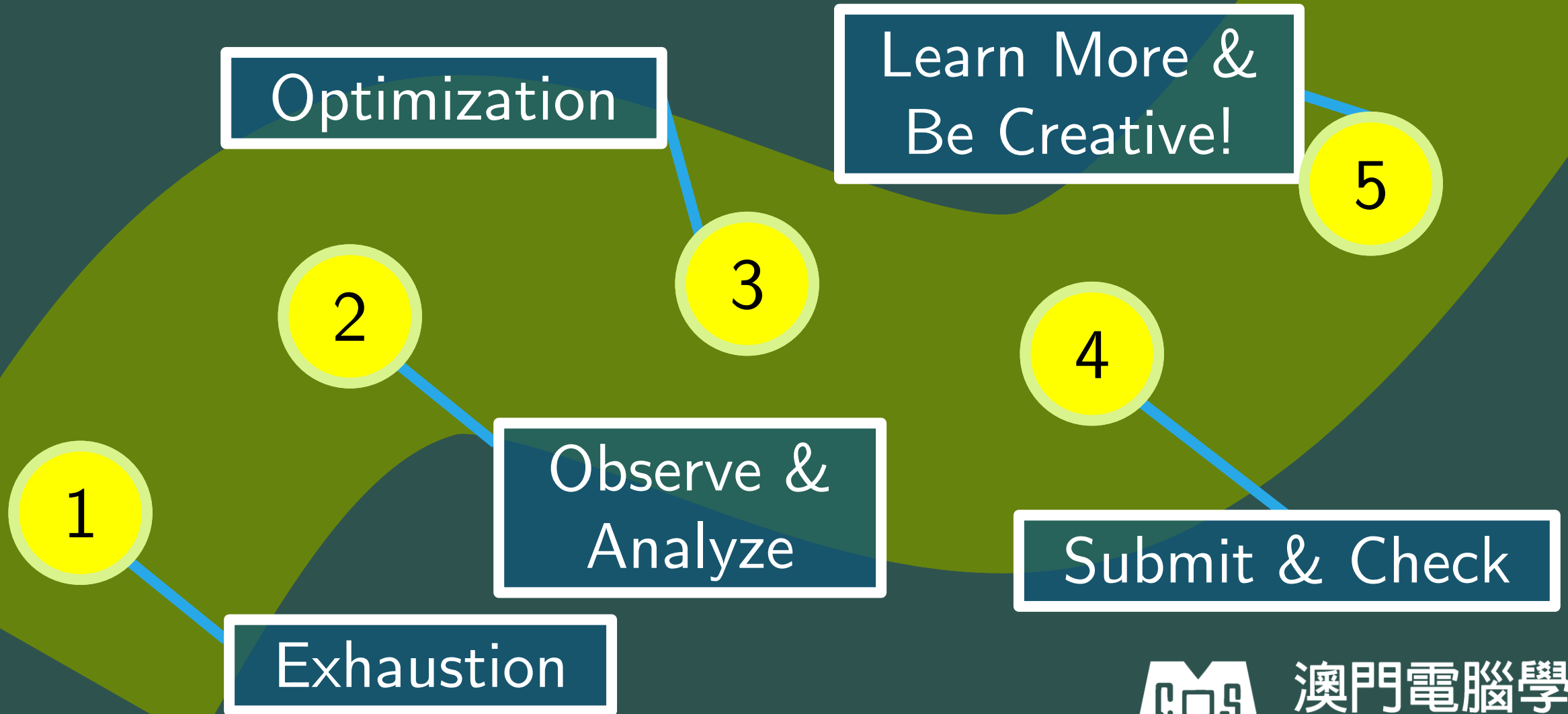
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What We Need to Do Next...

- **Combine** the two solutions together!
- **Determine B_0** (Hint: AM-GM Inequality).
- **Key Takeaway:** Identify the bottleneck; We can always try to find a proper threshold to divide a problem and solve them individually (and optimally)!



How Is the Problem Solved?



Homework

- **Some other problems on Harmonic Series**

洛谷 P2926 Patting Heads S

[NOIP2021] 报数

洛谷 P5190 [COCI2009-2010#5] PROGRAM

- **More on Finding Thresholds for Proper Optimization**

PCMS Contest - Prime Maze (Proposed by the Speaker)

AtCoder Beginner Contest Round 259 Problem Ex



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Thank you :-)

- Any questions?
- Hope to see your submission!
- Problem link:
<https://codeforces.com/gym/103176/problem/F>
- We do not suggest having puppet accounts as it violates CodeForces rules~

